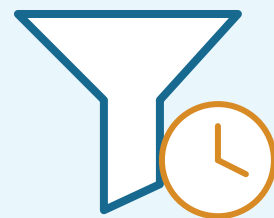


1. Enrolment roster and raw sources



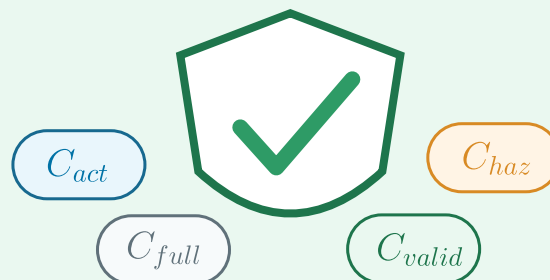
- Fixed roster $\mathcal{U}$
- Registration R_i and withdrawal W_i
- LMS and assessment events
- No event-based row deletion

2. Cutoff-first temporal filtering



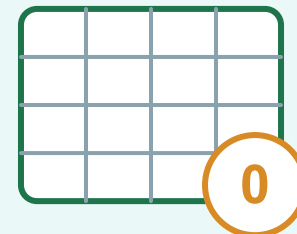
- Truncate before joins
- Keep records available at t
- Train-only preprocessing
- Temporal provenance audit

3. Eligibility resolution



- Activity state $A_i(t)$
- Eligibility: $R_i < t$, $W_i \geq t$
- Emit protocol masks
- Masks, not row deletion

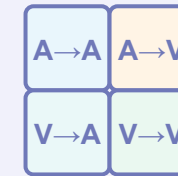
4. Roster-first representation



- Left join evidence on $\mathcal{U}$
- Structural zeros for silence
- Explicit no_activity=1
- One aligned $X^{(t)}$

5. Matched 2x2 experimental design

Train × score



A: activity-conditioned V: cutoff-valid

- Train on A and V
- Score on A and V
- Shared grouped folds
- Matched model heads

6. Metrics, decisions, and inference



- Population diagnostics
- ROC/PR, Brier, and ECE
- Fixed-budget top-k lists
- Paired cluster-bootstrap intervals

Stages 1 - 4 preserve $\mathcal{U}$: every protocol is a mask over one common representation.

Outputs: decomposition, valid intervention lists, and uncertainty.